BIOL 250 — Lower Extremity Quick-Reference Notes

Open-note speed reference — built from GR I–V answer choices + full Martini textbook sections (pelvic girdle & lower limb bones, hip/knee/ankle joints, pelvic & lower limb muscles, pelvic & lower limb vessels)

QUICK-SCAN INDEX

Alphabetical fast-lookup — for 15–25 sec/question recall. Compressed facts only; see full sections for detail.

Term	Quick fact / answer
Acetabular labrum	Fibrocartilage rim deepening the acetabulum
Adductor brevis	Obturator n.; adducts + flexes thigh
Adductor longus	Obturator n.; adducts + flexes + medially rotates thigh
Adductor magnus	Obturator n. + tibial (sciatic) n. — dual innervation; ant. flexes, post. extends thigh
Anterior cruciate ligament (ACL)	Resists anterior tibial translation
Anterior talofibular ligament	Most commonly torn ligament in ankle inversion sprains
Biceps femoris, long head	Sciatic (tibial); extends hip, flexes + laterally rotates knee
Biceps femoris, short head	Sciatic (common fibular); flexes + laterally rotates knee only
Calcaneofibular ligament	Lateral ankle ligament, malleolus → calcaneus
Common fibular nerve	Fibular head injury → foot drop
Deep fibular nerve	Anterior compartment; foot drop, loss of dorsiflexion
Deltoid ligament	Medial ankle; resists eversion
Extensor digitorum longus	Deep fibular n.; extends toes, dorsiflexes ankle
Femoral nerve	Lumbar plexus; quadriceps/sartorius/iliacus/pectineus — loss = no knee extension
Fibularis (peroneus) longus/brevis	Superficial fibular n.; evert foot
Flexor retinaculum (ankle)	Medial ankle band forming the tarsal tunnel (tibial n., vessels, flexor tendons)
Gastrocnemius	Tibial n.; plantarflexes ankle, flexes knee
Gluteus maximus	Inferior gluteal n.; extends + laterally rotates hip
Gluteus medius/minimus	Superior gluteal n.; abduct + medially rotate hip
Gracilis	Obturator n.; adducts thigh, flexes + medially rotates knee
Hamstrings	Biceps femoris, semitendinosus, semimembranosus — origin = ischial tuberosity
Hip fractures	Femoral neck fx very dangerous in elderly — disrupt blood supply to femoral head, high complication/immobility risk
Iliacus	Femoral n.; flexes hip
Iliofemoral ligament	Strongest hip ligament; ilium → femur
Iliopsoas	Iliacus + psoas major combined, deep to inguinal ligament
Inferior extensor retinaculum	Y-shaped band over dorsum of foot, holds extensor tendons
Inguinal canal	Passage through lower abdominal wall — distinct from inguinal ligament; transmits spermatic cord/round ligament
Inguinal ligament	External oblique aponeurosis band, ASIS → pubic tubercle; iliopsoas + femoral vessels pass deep to it
Intertarsal joints	Permit flexion/extension AND abduction/adduction, not just gliding
Ischial tuberosity	Hamstring origin; weight-bearing when seated
Ischiofemoral ligament	Tenses on extension + medial rotation; limits hyperextension
Knee joint	Modified hinge; unlocked by popliteus to initiate flexion
Lateral femoral cutaneous n.	Sensory only; meralgia paresthetica
Lateral rotator group	Piriformis, obturator internus/externus, gemelli, quadratus femoris
Ligament of femoral head (ligamentum teres)	Tenses on FLEXION + EXTERNAL (lateral) rotation of thigh; carries artery to femoral head
Linea aspera	Femoral shaft ridge; muscle position (ant/med vs post/lat) determines flexor/extensor/adductor action
Lumbar plexus	T12–L4; anterior/medial thigh

Term	Quick fact / answer
Medial/lateral meniscus	C-shaped fibrocartilage knee pads; medial meniscus attached to tibial collateral ligament
Obturator nerve	Medial thigh adductors, gracilis; weak thigh adduction if injured
Patellar ligament	Patella → tibial tuberosity
Pectineus	Femoral n.; flexes + adducts hip
Piriformis	Sacral branches S1–S2; laterally rotates + abducts hip
Plantar fasciitis	Inflammation at calcaneal attachment of plantar aponeurosis
Popliteus	Tibial n.; unlocks/medially rotates knee to initiate flexion
Posterior cruciate ligament (PCL)	Resists posterior tibial translation
Psoas major	Lumbar plexus (L2–L3); flexes hip and/or lumbar spine
Pubofemoral ligament	Pubis → femur, anterior/inferior hip capsule
Pudendal nerve	S2–S4; supplies perineum
Quadratus femoris	Sacral plexus n.; laterally rotates hip
Quadriceps femoris	Femoral n.; extends knee; common tendon → patella → patellar ligament
Rectus femoris	Femoral n.; only quad crossing hip — flexes hip + extends knee
Retinacula (ankle)	Superior extensor (above ankle ant.), inferior extensor (Y-shaped, dorsum), flexor (medial, tarsal tunnel)
Sartorius	Femoral n.; flexes hip + knee, laterally rotates thigh
Sciatic nerve	Largest nerve in body; splits into tibial + common fibular nn.
Semitendinosus/semimembranosus	Sciatic (tibial); extend hip, flex + medially rotate knee
Soleus	Tibial n.; plantarflexes ankle, deep to gastrocnemius
Subtalar joint	Talus ↔ calcaneus; inversion/eversion
Superficial fibular nerve	Lateral compartment + dorsum sensation
Superior extensor retinaculum	Band just above ankle, anteriorly, holds extensor tendons
Superior/inferior gluteal nerve	Sup. = medius/minimus/TFL (Trendelenburg gait); Inf. = maximus (weak hip extension)
Tensor fasciae latae (TFL)	Superior gluteal n.; abducts + medially rotates hip via IT tract
Tibial nerve	Posterior leg + sole of foot; loss = no plantarflexion
Tibialis anterior	Deep fibular n.; dorsiflexes + inverts foot
Tibialis posterior	Tibial n.; plantarflexes + inverts foot
Tibiofibular joint (distal)	Syndesmosis (fibrous) — like radius-ulna interosseous membrane
Tom Dick And [Nervous] Harry	Order behind medial malleolus: Tib. posterior, flex. Digitorum longus, tibial Artery, tibial Nerve, flex. Hallucis longus

1. Muscles — Action & Innervation

1.1 Gluteal / hip movers

Muscle	Origin → Insertion	Innervation	Action
Gluteus maximus	Iliac crest, post. gluteal line, sacrum/coccyx → iliotibial tract + gluteal tuberosity	Inferior gluteal n. (L5–S2)	Extend + laterally rotate hip; abducts (superior fibers)
Gluteus medius	Ilium between post./ant. gluteal lines → greater trochanter	Superior gluteal n. (L4–S1)	Abduct + medially rotate hip; stabilizes pelvis in stance
Gluteus minimus	Ilium between inf./ant. gluteal lines → greater trochanter	Superior gluteal n. (L4–S1)	Abduct + medially rotate hip
Tensor fasciae latae (TFL)	Iliac crest + ASIS → iliotibial tract	Superior gluteal n. (L4–S1)	Knee extension/lateral rotation via IT tract; abducts + medially rotates hip

Muscle	Origin → Insertion	Innervation	Action
Piriformis	Anterolateral sacrum → greater trochanter	Branches of S1–S2	Lateral rotation + abduction of hip
Obturator externus	Margins of obturator foramen → trochanteric fossa	Obturator n. (L3–L4)	Lateral rotation of hip
Obturator internus	Margins of obturator foramen → greater trochanter	Special sacral plexus n. (L5–S2)	Lateral rotation + abduction of hip
Gemelli (sup./inf.)	Ischial spine / ischial tuberosity → greater trochanter via obturator internus tendon	N. to obturator internus / n. to quadratus femoris	Lateral rotation + abduction of hip
Quadratus femoris	Lateral border of ischial tuberosity → intertrochanteric crest	Special sacral plexus n. (L4–S1)	Lateral rotation of hip
Iliacus	Iliac fossa → femur distal to lesser trochanter (fused w/ psoas major tendon)	Femoral n. (L2–L3)	Flexes hip
Psoas major	T12–L5 vertebrae → lesser trochanter	Lumbar plexus branches (L2–L3)	Flexes hip and/or lumbar spine

Iliacus + psoas major together = "iliopsoas"; both pass deep to the inguinal ligament. Piriformis/obturators/gemelli/quadratus femoris = "lateral rotator group" — dominant rotators are piriformis, obturator externus, obturator internus.

Don't confuse: the inguinal ligament is the band formed by the external oblique aponeurosis (ASIS → pubic tubercle), while the inguinal canal is a separate structure — a passage/tunnel through the lower abdominal wall (running parallel just above/deep to the inguinal ligament) that transmits the spermatic cord (M) or round ligament of uterus (F) plus ilioinguinal n.

1.2 Anterior & medial thigh

Muscle	Origin → Insertion	Innervation	Action
Sartorius	ASIS → medial proximal tibia (pes anserinus)	Femoral n. (L2–L3)	Flex hip + knee, laterally rotate thigh ('tailor's muscle'); originates at ASIS, inserts near tibial tuberosity
Rectus femoris	AIIS + ilium above acetabulum → tibial tuberosity via quadriceps tendon/patella	Femoral n. (L2–L4)	Flex hip + extend knee (only quad crossing hip)
Vastus lateralis / medialis / intermedius	Femoral shaft + linea aspera (lateral/medial lips) → tibial tuberosity via quadriceps tendon/patella	Femoral n. (L2–L4)	Extend knee
Quadriceps femoris (group of above 4)	Combined origins of all 4 heads → tibial tuberosity via common quadriceps tendon → patella → patellar ligament	Femoral n. (L2–L4)	Knee extension; common quadriceps tendon → patella → patellar ligament → tibial tuberosity

Muscle	Origin → Insertion	Innervation	Action
Pectineus	Pectineal line of pubis (superior pubic ramus) → pectineal line of femur (below lesser trochanter)	Femoral n. (L2–L4)	Flex + adduct hip
Adductor longus	Anterior pubic body → linea aspera (middle third)	Obturator n. (L2–L4)	Adduct + flex + medially rotate thigh
Adductor brevis	Inferior pubic ramus → linea aspera (upper part) + pectineal line	Obturator n. (L2–L4)	Adduct + flex thigh
Adductor magnus	Ischiopubic ramus (adductor part) + ischial tuberosity (hamstring part) → linea aspera (adductor part) + adductor tubercle (hamstring part)	Obturator n. AND sciatic n. (tibial portion) (L2–L4, L4–S1)	Adduct thigh; anterior part flexes, posterior part extends thigh (dual innervation — only LE muscle with two separate nerve supplies)
Gracilis	Pubic body / inferior pubic ramus → pes anserinus on medial tibia	Obturator n. (L2–L4)	Adduct thigh, flex + medially rotate knee

1.3 Posterior thigh (hamstrings) & popliteus

Muscle	Origin → Insertion	Innervation	Action
Biceps femoris — long head	Ischial tuberosity → head of fibula	Sciatic n., tibial portion (S1–S3)	Extend hip, flex + laterally rotate knee
Biceps femoris — short head	Linea aspera (lateral lip) of femur → head of fibula	Sciatic n., common fibular branch (L5–S2)	Flex + laterally rotate knee (no hip action — doesn't cross hip)
Semitendinosus	Ischial tuberosity → pes anserinus on medial tibia	Sciatic n., tibial portion (L5–S2)	Extend hip, flex + medially rotate knee
Semimembranosus	Ischial tuberosity → posteromedial tibial condyle	Sciatic n., tibial portion (L5–S2)	Extend hip, flex + medially rotate knee
Popliteus	Lateral femoral condyle → posterior tibia (above soleal line)	Tibial n. (L4–S1)	Unlocks/medially rotates knee (laterally rotates femur on tibia) to initiate flexion

Hamstrings = biceps femoris + semitendinosus + semimembranosus, all originate at ischial tuberosity (except short head of biceps femoris, which originates on linea aspera) and cross the knee posteriorly. Mnemonic for split innervation: only the short head of biceps femoris gets common fibular n. — everything else hamstring gets tibial n.

1.4 Leg (knee/ankle/foot movers)

Muscle	Origin → Insertion	Innervation	Action
Gastrocnemius	Medial + lateral femoral condyles → calcaneus via calcaneal (Achilles) tendon	Tibial n. (S1–S2)	Plantarflex ankle, flex knee; originates from femoral condyles (medial + lateral heads); fabella = sesamoid bone sometimes found in lateral head tendon
Soleus	Head/posterior fibula + soleal line of	Tibial n. (S1–S2)	Plantarflex ankle

Muscle	Origin → Insertion	Innervation	Action
	tibia → calcaneus via calcaneal tendon		(deep to gastrocnemius); + gastrocnemius = "triceps surae" → calcaneal (Achilles) tendon
Plantaris	Lateral femoral condyle (above gastrocnemius) → calcaneus, blending into calcaneal tendon	Tibial n. (L4–S1)	Weak plantarflexion/knee flexion
Tibialis posterior	Posterior tibia/fibula + interosseous membrane → navicular + other tarsals	Tibial n. (L5–S1)	Plantarflex + invert foot (deep posterior compartment)
Flexor digitorum longus	Posterior tibia → distal phalanges of toes 2–5	Tibial n. (L5–S1)	Flex toes (deep posterior compartment)
Flexor hallucis longus	Posterior fibula → distal phalanx of great toe	Tibial n. (L5–S1)	Flex great toe (deep posterior compartment)
Tibialis anterior	Lateral tibial condyle/shaft → medial cuneiform + base of 1st metatarsal	Deep fibular n. (L4–S1)	Dorsiflex + invert foot
Extensor digitorum longus	Lateral tibial condyle + fibula → middle/distal phalanges of toes 2–5 (extensor expansion)	Deep fibular n. (L5–S1)	Extend toes, dorsiflex ankle
Extensor hallucis longus	Middle fibula + interosseous membrane → distal phalanx of great toe	Deep fibular n. (L5–S1)	Extend great toe, dorsiflex ankle
Fibularis (peroneus) longus	Head/proximal fibula → medial cuneiform + base of 1st metatarsal	Superficial fibular n. (L4–S1)	Evert foot, weak plantarflex (lateral compartment); tendon crosses sole to support transverse arch
Fibularis (peroneus) brevis	Distal fibula → base of 5th metatarsal	Superficial fibular n. (L4–S1)	Evert foot, weak plantarflex (lateral compartment)
Fibularis (peroneus) tertius	Distal fibula → base of 5th metatarsal (dorsal surface)	Deep fibular n. (L4–S1)	Dorsiflex + evert foot

Compartment rule: anterior leg = deep fibular n.; lateral leg = superficial fibular n.; posterior leg (superficial+deep) = tibial n. Tibial n. innervates the deep posterior compartment specifically (per GR V). "Tom, Dick, And [Nervous] Harry" = structures passing posterior to medial malleolus in order: Tibialis posterior tendon, flexor Digitorum longus tendon, posterior tibial Artery, tibial Nerve, flexor Hallucis longus tendon.

1.5 Intrinsic foot muscles

Muscle	Origin → Insertion	Innervation	Action
Extensor digitorum brevis	Calcaneus (dorsal/lateral surface) → extensor tendons of toes 2–4	Deep fibular n.	Extend toes (only intrinsic muscle on dorsum of foot)
Flexor digitorum brevis	Calcaneal tuberosity (plantar surface) → middle phalanges of toes 2–5	Medial plantar n. (branch of tibial n.)	Flex toes 2–5 (middle phalanges)
Quadratus plantae	Calcaneus → tendon of flexor digitorum longus	Lateral plantar n.	Assists flexor digitorum longus; origin = calcaneus
Abductor hallucis	Calcaneus (medial tubercle) → proximal phalanx of great toe (medial side)	Medial plantar n.	Abduct + flex great toe; forms medial border of foot
Flexor hallucis brevis	Cuboid + lateral cuneiform → proximal phalanx of great toe	Medial plantar n.	Flex great toe
Adductor hallucis (oblique + transverse heads)	Oblique head: metatarsals 2–4; Transverse head: MTP joint ligaments	Lateral plantar n.	Adduct great toe; transverse head

Muscle	Origin → Insertion	Innervation	Action
	→ proximal phalanx of great toe (lateral side)		helps maintain transverse arch
Abductor digiti minimi	Calcaneus → proximal phalanx of little toe	Lateral plantar n.	Abduct + flex little toe; origin = calcaneus
Flexor digiti minimi brevis	Base of 5th metatarsal → proximal phalanx of little toe	Lateral plantar n.	Flex little toe
Lumbricals (4)	Tendons of flexor digitorum longus → extensor expansions of toes 2–5	Medial + lateral plantar nn.	Flex MTP, extend IP joints (like hand lumbricals)
Plantar / dorsal interossei	Metatarsals → proximal phalanges (via extensor expansions)	Lateral plantar n.	Adduct/abduct toes (PAD/DAB at toes, same logic as hand)

All originate primarily on tarsal + metatarsal bones; support the longitudinal + transverse arches and fine-tune toe movement. Sole muscles = medial + lateral plantar nn. (tibial n. branches); dorsum (extensor digitorum brevis) = deep fibular n. Plantar fasciitis/heel spur = inflammation/bony projection at calcaneal attachment of the plantar aponeurosis (deep fascia supporting the longitudinal arch).

2. Nerves — Lumbar & Sacral Plexus

Lumbar plexus = T12–L4 (anterior thigh/medial thigh). Sacral plexus = L4–S4 (posterior thigh/leg/foot). Together often called the lumbosacral plexus.

Nerve	Plexus	Supplies	Injury clue
Femoral nerve	Lumbar (L2–L4)	Anterior thigh: quadriceps, sartorius, iliacus, pectineus	Loss of knee extension
Obturator nerve	Lumbar (L2–L4)	Medial thigh adductors, gracilis	Weak thigh adduction
Lateral femoral cutaneous n.	Lumbar (L2–L3)	Sensory only, anterolateral thigh	Meralgia paresthetica (tingling thigh)
Iliohypogastric / ilioinguinal n.	Lumbar (L1)	Abdominal wall, groin, genital region	—
Sciatic nerve	Sacral (L4–S3)	Posterior thigh (hamstrings) → splits into tibial + common fibular	Largest nerve in body; sciatica
Tibial nerve	Sacral (from sciatic)	Posterior leg (gastrocnemius, soleus, tibialis posterior, toe flexors), sole of foot	Loss of plantarflexion
Common fibular (peroneal) n.	Sacral (from sciatic)	Splits to superficial + deep fibular n.	Fibular head injury → foot drop
Superficial fibular n.	branch of common fibular	Lateral compartment (fibularis longus/brevis) + dorsum sensation	—
Deep fibular n.	branch of common fibular	Anterior compartment (tibialis anterior, extensors)	Foot drop, loss of dorsiflexion
Superior gluteal n.	Sacral (L4–S1)	Gluteus medius, minimus, tensor fasciae latae	Trendelenburg gait (pelvis drop)
Inferior gluteal n.	Sacral (L5–S2)	Gluteus maximus	Weak hip extension
Pudendal nerve	Sacral (S2–S4)	Perineum	—

3. Bones & Landmarks (one line each)

3.1 Pelvis

Term	Quick ID
Ilium / ischium / pubis	3 bones fusing to form each hip bone (os coxae)
Ala of ilium	Broad, flared wing-shaped portion of the ilium
Iliac crest	Superior rim of ilium — palpable at waistline
Iliac fossa	Concave internal (medial) surface of ilium
Iliac tuberosity	Roughened area on ilium posterior to iliac fossa — ligament attachment
Auricular surface	Ear-shaped articular surface of ilium — meets sacrum at sacroiliac joint

Term	Quick ID
Anterior / posterior / inferior gluteal lines	Curved ridges on lateral ilium surface marking gluteal muscle origins
Anterior superior iliac spine (ASIS)	Anterior iliac crest landmark; sartorius + inguinal ligament attach
Anterior inferior iliac spine (AIIS)	Below ASIS; rectus femoris attaches
Posterior superior iliac spine (PSIS)	Posterior iliac crest landmark
Posterior inferior iliac spine (PIIS)	Below PSIS, above greater sciatic notch
Greater / lesser sciatic notch	Posterior ischium/ilium gaps — sciatic n. + vessels pass through greater notch
Ischial tuberosity	Bears body weight when seated; hamstring origin
Ischial spine	Separates greater/lesser sciatic notches
Ischial ramus	Connects ischial tuberosity to the pubis, helps form obturator foramen
Acetabulum	Cup-shaped socket of hip joint (ilium+ischium+pubis); has lunate (articular) surface + acetabular fossa + acetabular notch
Lunate surface	Crescent-shaped articular portion lining the acetabulum
Acetabular notch/fossa	Gap/depression in acetabulum floor — non-articular, holds fat pad + ligament of femoral head
Obturator foramen	Large opening, ischium+pubis — obturator nerve/vessels pass through obturator groove
Obturator groove	Groove along pubis margin of obturator foramen — guides obturator vessels/nerve
Pubic ramus (sup./inf.) / tubercle / crest	Pubis landmarks — rami help form obturator foramen; crest/tubercle = anterior attachment sites
Pubic symphysis	Cartilaginous (amphiarthrotic) midline joint between the two pubic bones
Pectineal line	Ridge on superior pubic ramus — pectineus origin, part of pelvic brim
Arcuate line	Internal ridge of ilium, part of pelvic brim
Pelvic brim / inlet / outlet	Pelvic brim = boundary between greater (false) and lesser (true) pelvis; inlet = superior opening of true pelvis; outlet = inferior opening
Greater (false) vs. lesser (true) pelvis	Greater pelvis = above brim, bounded by ilium (abdominal); lesser pelvis = below brim, the true pelvic cavity
Perineum	Region inferior to the pelvic floor, between thighs, containing external genitalia + anus
Sacral promontory	Anterior lip of S1 — pelvic brim landmark
Median sacral crest / sacral foramina	Posterior sacrum landmarks — fused spinous processes / nerve exit points

3.1b Male vs. female pelvis (Table 7.1)

Feature	Male	Female
Pelvic inlet shape	Heart-shaped, narrower	Oval/rounder, wider
Iliac fossa	Deeper	Shallower
Ilium orientation	More vertical	Less vertical, flares laterally
Subpubic angle	Narrower (<90°)	Wider (>90°)
Acetabula	Closer together, face more laterally	Farther apart, face more anteriorly
Obturator foramen	Oval	Triangular
Ischial spines	Point more medially	Point more laterally — wider pelvic outlet
Sacrum / coccyx	Narrower, projects less, points more anteriorly, more fused vertebrae	Wider, shorter, straighter — accommodates childbirth

3.2 Femur

Term	Quick ID
Head	Proximal ball ↔ acetabulum
Fovea capitis (fovea for ligament of head)	Small pit on head — attachment of ligament of femoral head
Neck	Connects head to shaft; common fracture site ('hip fracture')
Greater trochanter	Lateral proximal projection — gluteal muscle attachment, palpable hip landmark
Lesser trochanter	Medial proximal projection — iliopsoas insertion
Intertrochanteric line / crest	Anterior/posterior ridges connecting the trochanters
Linea aspera	Posterior shaft ridge — adductor muscle attachments; muscle position relative to it (anterior/medial vs. posterior/lateral) determines flexor/extensor/adductor action at the thigh

Term	Quick ID
Gluteal tuberosity	Lateral ridge near linea aspera — gluteus maximus insertion
Pectineal line (femur)	Ridge running from lesser trochanter toward linea aspera — pectineus insertion
Medial / lateral supracondylar ridge	Ridges proximal to the condyles, diverging from linea aspera
Adductor tubercle	Small projection on medial epicondyle — adductor magnus insertion
Popliteal surface	Flat triangular area on posterior distal femur, floor of popliteal fossa
Medial / lateral condyle	Distal rounded ends ↔ tibia, form knee joint
Medial / lateral epicondyle	Bony bumps above condyles — collateral ligament attachments
Intercondylar (fossa/notch)	Posterior groove between the two condyles

Hip fractures (esp. femoral neck fractures) are among the most dangerous fractures in elderly people — they disrupt blood supply (medial/lateral circumflex femoral aa.) to the femoral head, with high risk of avascular necrosis, complications, and prolonged immobility.

3.3 Tibia, fibula, patella, foot bones

Term	Quick ID
Tibial tuberosity	Anterior proximal tibia — patellar ligament insertion
Intercondylar eminence + medial/lateral intercondylar tubercles	Raised area between tibial condyles — cruciate ligament + meniscus attachment
Anterior margin (border)	Sharp anterior ridge of tibial shaft — "shin"
Interosseous border (tibia)	Lateral tibial shaft ridge — anchors crural interosseous membrane
Soleal line	Diagonal ridge on posterior proximal tibia — soleus origin
Medial malleolus	Distal medial tibia — inner ankle bump
Lateral malleolus	Distal fibula — outer ankle bump
Head / neck of fibula	Proximal end — lateral collateral ligament + biceps femoris tendon attach to head
Interosseous border (fibula)	Medial fibular shaft ridge — anchors crural interosseous membrane
Crural interosseous membrane	Connects tibia + fibula shafts (like the radius/ulna membrane in the forearm)
Patella — base / apex	Base = broad superior border; apex = pointed inferior tip (patellar ligament attaches at apex)
Patella — medial/lateral articular facets	Posterior surface facets articulating with the femoral condyles
Talus	Largest tarsal; trochlea articulates with tibia/fibula at ankle
Calcaneus	Heel bone; calcaneal (Achilles) tendon inserts here; medial tubercle = plantar fascia/intrinsic muscle origin
Navicular, cuboid, medial/intermediate/lateral cuneiform	Remaining tarsal bones, midfoot
Metatarsals I–V / phalanges	Same pattern as hand: 2 phalanges great toe, 3 each other toe
Longitudinal arch	Runs heel-to-toe (medial portion higher than lateral); distributes weight front-to-back
Transverse arch	Runs side-to-side across the midfoot (tarsals/metatarsals); fibularis longus + tibialis posterior tendons help support it

3.4 Surface anatomy / regional terms

Term	Region
Femur(al)	Thigh
Crus / sura	Leg (crus=anterior leg term, sura=calf)
Patella	Knee region (anterior)
Popliteus / popliteal	Back of knee
Tarsus	Ankle
Pes	Foot (whole)
Planta	Sole of foot
Hallux	Big toe
Calcaneus	Heel (bone + region)
Inguen	Groin
Lumbus	Lower back/loin

Term	Region
Gluteus	Buttock
Pubis	Pubic region

4. Joints & Ligaments

Joint	Type	Notes
Sacroiliac (SI) joint	Synovial plane (gliding) diarthrosis	Sacrum ↔ ilium auricular surfaces; very little movement, heavily ligament-reinforced
Pubic symphysis	Amphiarthrosis (cartilaginous, fibrocartilage pad)	Slight movement; can widen slightly during childbirth
Hip (coxal)	Ball-and-socket, triaxial	Deep, stable socket (vs. shallow shoulder) — rarely dislocates
Knee	Hinge + gliding (condylar) — modified hinge	Most complex joint; extension preceded by 'unlocking' rotation (popliteus)
Tibiofibular (proximal)	Synovial plane/gliding joint	—
Tibiofibular (distal)	Syndesmosis (fibrous)	Same classification as radius-ulna interosseous membrane; stabilized by tibiofibular ligaments
Ankle (tibiotalar)	Hinge	Major weight-bearing ankle articulation; tibia+fibula malleoli grip the talus
Subtalar joint	Synovial plane (gliding)	Talus ↔ calcaneus — inversion/eversion
Talonavicular joint	Synovial plane (gliding)	Part of the transverse tarsal joint
Calcaneocuboid joint	Synovial plane (gliding)	Part of the transverse tarsal joint
Cuneonavicular / intertarsal joints	Synovial plane (gliding)	Permit flexion/extension AND abduction/adduction between remaining tarsals, not just gliding
Tarsometatarsal / MTP / IP joints of foot	Plane / condylar / hinge	MTP+IP permit flexion/extension + some ab/adduction at MTP

Ligament	Connects / notes
Iliofemoral ligament	Ilium → femur, anterior hip capsule — strongest hip ligament
Pubofemoral ligament	Pubis → femur, anterior/inferior hip capsule
Ischiofemoral ligament	Ischium → femur, posterior hip capsule; tenses on extension + medial rotation, limits hyperextension
Transverse acetabular ligament	Crosses the acetabular notch, completing the acetabular rim
Acetabular labrum	Fibrocartilage rim deepening the acetabulum
Ligament of the femoral head (ligamentum teres)	Fovea capitis → acetabulum; carries a small artery to femoral head; tenses on flexion + external (lateral) rotation of the thigh
Anterior cruciate ligament (ACL)	Tibia (intercondylar area) → femur, crosses anteriorly; resists anterior tibial translation
Posterior cruciate ligament (PCL)	Tibia → femur, crosses posteriorly; resists posterior tibial translation
Tibial (medial) collateral ligament	Femur medial epicondyle → tibia; resists valgus stress; attached to medial meniscus
Fibular (lateral) collateral ligament	Femur lateral epicondyle → fibula head; resists varus stress
Patellar ligament	Patella → tibial tuberosity (continuation of quadriceps tendon)
Patellar retinaculae (medial/lateral)	Fibrous bands flanking the patella, reinforcing the joint capsule
Popliteal ligaments (oblique/arcuate)	Reinforce the posterior knee joint capsule
Medial / lateral meniscus	C-shaped fibrocartilage pads cushioning femoral/tibial condyles; medial meniscus attached to tibial collateral ligament
Suprapatellar / prepatellar / infrapatellar bursae	Fluid-filled sacs reducing friction around the patella/patellar ligament
Deltoid ligament (medial ankle)	Broad ligament, medial malleolus → talus/calcaneus/navicular; resists eversion
Anterior talofibular ligament	Lateral ankle; lateral malleolus → talus; most commonly torn in inversion sprains
Posterior talofibular ligament	Lateral ankle; lateral malleolus → talus, posterior
Calcaneofibular ligament	Lateral ankle; lateral malleolus → calcaneus
Anterior / posterior tibiofibular ligaments	Stabilize the distal tibiofibular syndesmosis

Lateral ankle (deltoid is medial) has 3 named ligaments — anterior talofibular, posterior talofibular, calcaneofibular — all resisting inversion; inversion injuries are far more common than eversion injuries because the lateral ligaments are weaker than the deltoid. A bimalleolar (Pott's) fracture involves both malleoli.

4.1 Retinacula of the ankle

Retinaculum	Location / function
Superior extensor retinaculum	Broad band just above the ankle, anteriorly — holds extensor tendons (tibialis anterior, EDL, EHL) against the leg
Inferior extensor retinaculum	Y-shaped band over the dorsum of the foot — continues holding extensor tendons down distal to the ankle
Flexor retinaculum (ankle)	Spans the medial ankle (medial malleolus → calcaneus) — forms the tarsal tunnel for tibial n., posterior tibial vessels, and flexor tendons

5. Vessels

Arterial path: common iliac a. → internal iliac a. (pelvis/gluteal region) + external iliac a. → femoral a. (thigh) → deep femoral a. + medial/lateral circumflex femoral aa. branch off → femoral a. continues → popliteal a. (behind knee, gives off descending genicular a.) → splits into anterior tibial a. (→ dorsalis pedis a. on dorsum of foot) + posterior tibial a. (→ fibular/peroneal a. + medial/lateral plantar aa. on sole). Dorsal arch (from dorsalis pedis) and plantar arch (from plantar aa.) anastomose in the foot.

Vessel	Notes
Internal iliac artery	Supplies pelvis/gluteal region: superior + inferior gluteal aa., internal pudendal a., obturator a., lateral sacral aa.
External iliac artery	Continues as femoral a. after passing under inguinal ligament
Femoral artery	Continuation of external iliac a., anterior thigh, in femoral triangle
Deep femoral (profunda femoris) artery	Major branch of femoral a.; supplies deep/posterior thigh muscles
Medial / lateral circumflex femoral arteries	Branches of deep femoral a.; wrap around femoral neck — major blood supply to femoral head/neck (injury risk in hip fracture)
Descending genicular artery	Branches off femoral a. near knee, contributes to genicular anastomosis around the knee
Popliteal artery	Femoral a. continues behind the knee (popliteal fossa)
Anterior tibial artery	→ dorsalis pedis a.; palpated on dorsum of foot
Posterior tibial artery	Palpated inferomedial to medial malleolus; → fibular (peroneal) a. + medial/lateral plantar aa.
Fibular (peroneal) artery	Branch of posterior tibial a.; supplies lateral leg/fibula region
Dorsalis pedis artery	Palpated anteromedial/dorsal surface of foot; feeds the dorsal arch
Great saphenous vein	Longest vein in body; superficial, medial leg/thigh → femoral vein
Small saphenous vein	Superficial, posterior calf → popliteal vein
Dorsal / plantar venous arches	Superficial foot veins feeding the saphenous system
Anterior / posterior tibial veins, fibular vein	Deep veins paralleling the corresponding arteries → popliteal v.
Popliteal vein	Behind knee, deep, parallels popliteal a.
Femoral vein	Receives deep femoral v. + femoral circumflex v. → becomes external iliac v.
Internal iliac vein	Tributaries: gluteal, internal pudendal, obturator, lateral sacral veins
Common iliac vein → inferior vena cava (IVC)	External + internal iliac vv. join → common iliac v. → IVC

6. Other Real Terms That Show Up as Answer Choices

Cross-topic distractors seen in LE GRs (lymphatic system figure, bone classification figure). One line each so you can rule them out fast.

Term	Quick ID
Inguinal lymph nodes	Groin region — drain the lower limb
Lumbar lymph nodes	Posterior abdominal wall, along aorta
Cervical lymph nodes	Neck region
Cisterna chyli	Dilated sac at the start of the thoracic duct (L1–L2 level), collects lymph from lower body
Thoracic duct	Largest lymphatic vessel; drains most of body into left subclavian vein
Pneumatized bone	Contains air-filled sinuses (e.g., skull bones) — not a leg bone classification
Sesamoid bone	Forms within a tendon (patella is the classic example)

Term	Quick ID
Synchondrosis / synostosis / gomphosis / suture	Other joint classifications: cartilage-joined / fused-by-bone / tooth-socket / interlocking skull seam — distractors for the tibiofibular syndesmosis question